

JESS Thermodynamic Database v8.9

Promethium

24-Jan-23

Reaction No. 2677							
F-1 + Pm+3 = Pm+3_F-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.63(3SF)	3	CRV	8781
2	25	0	Inf. Dilution	lgK:3.96(3SF)	4	EDH	8917
3	25	0.1	NaClO ₄	lgK:3.32(3SF)	5	MTP	6681
4	25	0.1	Unknown	lgK:3.52(3SF)	6	CRV	552
5	25	0.68	NaClO ₄	lgK:3.07(3SF)	5	EFE	8917
6	25	1	NaCl	lgK:3.11(0.02SD)	5	MDS	5385
7	25	1	NaClO ₄	lgK:3.2(0.05SD)	6	MDS	437

Reaction No. 5479							
EDTA-4 + Pm+3 = Pm+3_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NH ₄ Cl	lgK:17.0(0.04SD)	7	EIU	903
2	20	0.1	NH ₄ Cl	lgK:16.94(4SF)	6	MSL	906
3	20	0.1	NH ₄ ClO ₄	lgK:17.0(0.04SD)	5	MDS	903
4	23	0	NH ₄ ClO ₄	lgK:19.41(4SF)	4	MCE	906
5	23	0.1	NH ₄ ClO ₄	lgK:16.75(4SF)	4	MCE	906
6	25	0	Inf. Dilution	lgK:17.3(3SF)	1	ESC	

Reaction No. 6436							
Oxalic-2 + Pm+3 = Pm+3_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.01(4SF)	1	EOR	
2	25	0.1	Unknown	lgK:5.2(0.1SD)	4	MTP	6545

Reaction No. 6437							
Pm+3_Oxalic-2 + Oxalic-2 = Pm+3_Oxalic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:3.6(0.05SD)	4	MTP	6545

Reaction No. 6916							
Pm+3 + H+1_Gly-1 = Pm+3_H+1_Gly-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	2	NaClO4	lgK:0.4(0.05SD)	6	MDS	3232
2	11	2	NaClO4	lgK:0.5(0.08SD)	6	MDS	3232
3	25	2	NaClO4	lgK:0.7(0.03SD)	6	MDS	3232
4	40	2	NaClO4	lgK:0.8(0.03SD)	6	MDS	3232
5	25	0.1	Unknown	dH:4.(1SF)	3	CRV	552
6	25	2	NaClO4	dH:4.(0.4SD)	5	MDS	3232

Reaction No. 7555							
CO3-2 + Pm+3 = Pm+3_CO3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.81(3SF)	4	EFE	8712
2	25	0	Inf. Dilution	lgK:7.22(3SF)	3	CRV	8781

Reaction No. 9159							
NTA-3 + Pm+3 = Pm+3_NTA-3							
Note: Value of GugC fixed by similarity of this parameter with other REEs							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:11.(2SF)	3	MDS	1118
2	25	0	Inf. Dilution	lgK:12.8(3SF)	4	EES	
Note (2): Based on lanthanide neighbours (see Li & Byrne, Aq. Geochem., 1997)							
3	25	0.1	K+1 salt	lgK:19.7(3SF)	0	CRV	552

Reaction No. 9160							
Pm+3_NTA-3 + NTA-3 = Pm+3_NTA-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:8.7(2SF)	4	CMN	1118
2	25	0	Inf. Dilution	lgK:8.9(2SF)	1	ESC	

Reaction No. 12179							
Citric-3 + Pm+3 = Pm+3_Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.905(4SF)	1	EOR	
2	25	0.1	NaCl	lgK:7.00(3SF)	6	MIX	999
3	25	0.1	Unknown	lgK:7.7(0.08SD)	4	MDG	6545

Reaction No. 12180							
Pm+3_Citric-3 + Citric-3 = Pm+3_Citric-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.6(2SF)	1	EDH	
2	25	0.1	NaCl	lgK:4.91(3SF)	2	MIX	11516
Note (2): Ohyoshi A et al., J. Inorg. Nucl. Chem., 1972, 34, 1955							
3	25	0.1	Unknown	lgK:3.2(0.2SD)	2	MDG	6545

Reaction No. 12181							
Pm+3_Citric-3 + H+1_Citric-3 = Pm+3_H+1_Citric-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.3(2SF)	1	ELC	
2	25	0.1	Unknown	lgK:2.4(0.1SD)	0	MDG	6545

Reaction No. 12182							
2<Citric-3> + Pm+3 = Pm+3_Citric-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.75(4SF)	1	EOR	
2	25	0.1	NaCl	lgK:11.91(4SF)	6	MIX	999
3	25	0.1	Unknown	lgK:3.20(3SF)	0	MDG	6545

Reaction No. 12562							
2<NTA-3> + Pm+3 = Pm+3_NTA-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:19.71(4SF)	5	MDS	1118
2	25	0	Inf. Dilution	lgK:20.0(3SF)	1	ESC	

Reaction No. 13501							
Mandelic-1 + Pm+3 = Pm+3_Mandelic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.2	NaClO4	lgK:2.32(3SF)	6	MIX	999

Reaction No. 13502							
2<Mandelic-1> + Pm+3 = Pm+3_Mandelic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.2	NaClO4	lgK:4.02(3SF)	6	MIX	999

Reaction No. 13503							
3<Mandelic-1> + Pm+3 = Pm+3_Mandelic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.2	NaClO4	lgK:5.11 (3SF)	6	MIX	999

Reaction No. 14634							
CDTA-4 + Pm+3 = Pm+3_CDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NH4Cl	lgK:18.17 (4SF)	6	MDS	999
2	25	0	Unknown	lgK:21.16 (4SF)	6	MIX	999
3	25	0.1	Unknown	lgK:18.50 (4SF)	6	MIX	999
4	80	0.05	Unknown	lgK:18.99 (4SF)	6	MIX	999
5	80	0.06	Unknown	lgK:19.01 (4SF)	6	MIX	999
6	80	0.07	Unknown	lgK:18.93 (4SF)	6	MIX	999
7	80	0.17	Unknown	lgK:17.83 (4SF)	6	MIX	999

Reaction No. 19485							
2<Oxalic-2> + Pm+3 = Pm+3_Oxalic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NH4Cl	lgK:8.3 (2SF)	4	MDS	906
2	25	0	Inf. Dilution	lgK:10.03 (4SF)	1	EOR	
3	25	0.1	Unknown	lgK:8.80 (3SF)	5	MTP	11516
Note (3): Stepanov A, Zhur. Neorg. Khim., 1971, 16, 2981							

Reaction No. 19486							
3<Oxalic-2> + Pm+3 = Pm+3_Oxalic-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NH4Cl	lgK:11.8 (3SF)	4	MDS	906
2	25	0	Inf. Dilution	lgK:12.1 (3SF)	1	ESC	

Reaction No. 19568							
Lactic-1 + Pm+3 = Pm+3_Lactic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	1.5	KCl	lgK:2.54 (3SF)	6	MSC	999
2	20	0.2	NaClO4	lgK:2.43 (3SF)	6	MIX	999

Reaction No. 19569							
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2<Lactic-1> + Pm+3 = Pm+3_Lactic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	1.5	KCl	lgK:4.24(3SF)	6	MSC	999
2	20	0.2	NaClO4	lgK:4.20(3SF)	6	MIX	999
3	25	0.1	Unknown	lgK:5.38(3SF)	5	MIX	11516
Note (3): Deelstra H et al., Bull. Soc. Chim. Belges, 1967, 76, 557							

Reaction No. 19570							
3<Lactic-1> + Pm+3 = Pm+3_Lactic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.2	NaClO4	lgK:5.35(3SF)	6	MIX	999

Reaction No. 20036							
2OH2MePropan-1 + Pm+3 = Pm+3_2OH2MePropan-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.2	NaClO4	lgK:2.63(3SF)	6	MIX	999
2	25	0.1	Unknown	lgK:2.79(3SF)	6	MTP	999

Reaction No. 20037							
2<2OH2MePropan-1> + Pm+3 = Pm+3_2OH2MePropan-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.2	NaClO4	lgK:4.78(3SF)	6	MIX	999
2	23			lgK:6.00(3SF)	0	MCE	906
3	25	0.1	Unknown	lgK:4.69(3SF)	6	MTP	999

Reaction No. 20038							
3<2OH2MePropan-1> + Pm+3 = Pm+3_2OH2MePropan-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.2	NaClO4	lgK:5.90(3SF)	6	MIX	999
2	25	0.1	Unknown	lgK:6.20(3SF)	6	MTP	999

Reaction No. 22385							
Tartaric-2 + Pm+3 = Pm+3_Tartaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NH4Cl	lgK:3.9(2SF)	4	MDS	999

Reaction No. 22386							
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2<Tartaric-2> + Pm+3 = Pm+3_Tartaric-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NH4Cl	lgK:6.8(2SF)	0	MDS	999
Note (1): Low wgt for internal consistency							
2	22	0.1	NaCl	lgK:5.81(3SF)	6	MSC	999

Reaction No. 22858							
Pm+3 + H+1_Citric-3 + Citric-3 = Pm+3_H+1_Citric-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	24:26	0.15	LiClO4	lgK:11.13(4SF)	6	MDS	999

Reaction No. 22866							
Pm+3 + H+1_Citric-3 = Pm+3_H+1_Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaCl	lgK:5.46(3SF)	6	MIX	999

Reaction No. 22867							
Pm+3_H+1_Citric-3 + H+1_Citric-3 = Pm+3_H+1(2)_Citric-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaCl	lgK:8.42(3SF)	6	MIX	999

Reaction No. 24764							
Pm+3 + H+1_CO3-2 = Pm+3_H+1_CO3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.79(3SF)	3	CRV	8781

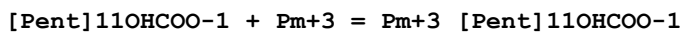
Reaction No. 27490							
Pm+3 + H+1_PhosAcet-3 = Pm+3_H+1_PhosAcet-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.15(3SF)	6	MIX	906

Reaction No. 27491							
Pm+3 + 2<H+1_PhosAcet-3> = Pm+3_H+1(2)_PhosAcet-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.50(3SF)	6	MIX	906

Reaction No. 27492							
Pm+3 + H+1(2)_PhosAcet-3 = Pm+3_H+1(2)_PhosAcet-3							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.75 (3SF)	6	MIX	906

Reaction No. 30236



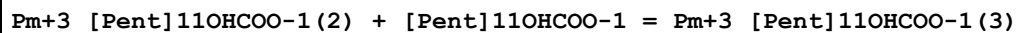
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:2.73 (3SF)	5	EIE	2150

Reaction No. 30237



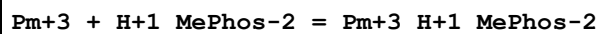
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:2.04 (3SF)	5	EIE	2150

Reaction No. 30238



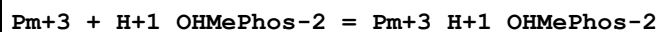
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:1.40 (3SF)	5	EIE	2150

Reaction No. 42828



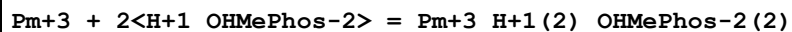
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.67 (3SF)	5	MIX	906
2	25	0.5	Many Media	lgK:1.7 (0.2SD)	6	CIU	7242

Reaction No. 42833



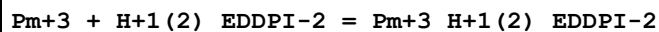
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.2	NH4ClO4	lgK:1.65 (3SF)	5	MIX	906

Reaction No. 42834



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.2	NH4ClO4	lgK:3.30 (3SF)	5	MIX	906

Reaction No. 43482



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	Unknown	lgK:5.78 (3SF)	4	MIX	906

Reaction No. 43494							
Pm+3 + H+1(2)_Etbis(ImMePhos)-4 = Pm+3_H+1(2)_Etbis(ImMePhos)-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NH4ClO4	lgK:5.7(0.06SD)	4	MIX	906

Reaction No. 44086							
HIMDA-2 + Pm+3 = Pm+3_HIMDA-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:9.0(0.05SD)	5	MDS	906

Reaction No. 44646							
Oxine-1 + Pm+3 = Pm+3_Oxine-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:6.91(3SF)	4	MDS	906

Reaction No. 44647							
2<Oxine-1> + Pm+3 = Pm+3_Oxine-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:13.25(4SF)	4	MDS	906

Reaction No. 44648							
3<Oxine-1> + Pm+3 = Pm+3_Oxine-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:19.08(4SF)	4	MDS	906

Reaction No. 52227							
Pm+3 + OH-1 = Pm+3_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.77(3SF)	3	EDH	8781

Reaction No. 52746							
IO3-1 + Pm+3 = Pm+3_IO3-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.81(3SF)	4	CRV	552
2	25	0.1	NaClO4	lgK:1.12(3SF)	5	MDS	5398

Reaction No. 52979							
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Cl-1 + Pm+3 = Pm+3_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15			lgK:0.70 (2SF)	0	MEP	5398
2	25	0	Inf. Dilution	lgK:0.31 (2SF)	3	CRV	8781
3	25	0.5	Inert	lgK:-0.48 (2SF)	4	MLC	8917
4	25	1	Inert	lgK:-0.54 (2SF)	4	MLC	8917

Reaction No. 52980							
Pm+3_Cl-1 + Cl-1 = Pm+3_Cl-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15			lgK:-0.8 (1SF)	0	MEP	5398

Reaction No. 53368							
Pm+3 + H+1 (2)_PO4-3 = Pm+3_H+1 (2)_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.51 (3SF)	4	MIX	5398
2	25	0	Inf. Dilution	lgK:2.27 (3SF)	3	CRV	8781
3	25	0.2	NH4ClO4	lgK:1.69 (3SF)	5	MCE	931

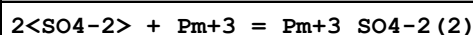
Reaction No. 53442							
P3O9-3 + Pm+3 = Pm+3_P3O9-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.74 (3SF)	4	MCE	931
2	25	0	Inf. Dilution	lgK:6.26 (3SF)	4	MIX	5398
3	25	0.2	NH4ClO4	lgK:3.80 (3SF)	4	MCE	931

Reaction No. 53830							
NO3-1 + Pm+3 = Pm+3_NO3-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.31 (3SF)	4	EOD	7141
2	25	0	Inf. Dilution	lgK:0.88 (2SF)	3	EDH	8781
3	25	1	HClO4	lgK:0.39 (2SF)	5	MDS	931
4	25	1	Inert	lgK:0.50 (2SF)	4	MLC	8917
5	25	1	NaClO4	lgK:0.26 (2SF)	5	MDS	931

Reaction No. 54039							
SO4-2 + Pm+3 = Pm+3_SO4-2							

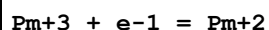
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	2	NaClO4	lgK:1.08 (3SF)	5	MDS	931
2	25	0	Inf. Dilution	lgK:3.5 (2SF)	3	EES	
3	25	0	Inf. Dilution	lgK:3.42 (4SF)	1	EOR	
4	25	0	Inf. Dilution	lgK:3.34 (3SF)	3	CRV	8781
5	25	2	NaClO4	lgK:1.34 (3SF)	5	MDS	931
6	40	2	NaClO4	lgK:1.49 (3SF)	5	MDS	931
7	55	2	NaClO4	lgK:1.60 (3SF)	5	MDS	931
8	0:55	2	NaClO4	dH:3.9 (2SF)	5	MTD	931

Reaction No. 54040



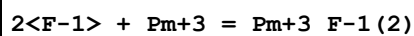
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	2	NaClO4	lgK:1.62 (3SF)	5	MDS	931
2	25	0	Inf. Dilution	lgK:5.1 (2SF)	3	EES	
3	25	0	Inf. Dilution	lgK:5.1 (4SF)	1	EOR	
4	25	2	NaClO4	lgK:1.88 (3SF)	5	MDS	931
5	40	2	NaClO4	lgK:2.00 (3SF)	5	MDS	931
6	55	2	NaClO4	lgK:2.28 (3SF)	5	MDS	931

Reaction No. 55662



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.6 (2SF)	0	CRV	6330
Note (1): Low wgt for inter-reaction consistency							
2	25	0	Inf. Dilution	E:-2.6 (2SF)	0	CRV	7761
3	25	0	Unknown	E:-2.5 (0.2SD)	3	ENS	5256
4	25	0	Inf. Dilution	dH:294.0 (4SF) kJ	3	CRV	7761

Reaction No. 57781



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.42 (3SF)	4	EDH	8917
2	25	0.68	NaClO4	lgK:5.08 (3SF)	5	EFE	8917
3	25	1	NaCl	lgK:5.60 (0.02SD)	6	MDS	5385

Reaction No. 65774

N3-1 + Pm+3 = Pm+3_N3-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	Unknown	lgK:0.52 (2SF)	6	CRV	552

Reaction No. 66000							
Pm+3_IO3-1(3)_(s) = Pm+3 + 3<IO3-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-11.3 (0.1SD)	4	CRV	552
2	25	0.1	Unknown	dH:5.8 (2SF)	6	CRV	552

Reaction No. 66252							
Pm+3 + 3<e-1> = Pm(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:663.(3SF) kJ	5	CRV	7421
2	25	0	Inf. Dilution	dG:158.000 (6SF)	2	CRV	9029
3	25	0	Inf. Dilution	E:-2.42 (3SF)	0	MDG	140
4	25	0	Inf. Dilution	E:-2.30 (3SF)	2	CRV	6330
5	25	0	Inf. Dilution	E:-2.29 (3SF)	4	ENS	7276
6	25	0	Inf. Dilution	E:-2.30 (3SF)	2	CRV	7761
7	25	0	Inf. Dilution	dH:690.8 (4SF) kJ	3	CRV	7761
8	25	0	Inf. Dilution	dH:164.000 (6SF)	3	CRV	9029

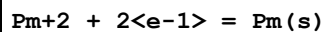
Reaction No. 66870							
Pm+3_OH-1(3)_(s) = Pm+3 + 3<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-24.0 (3SF)	4	EES	
2	25	0	Inf. Dilution	lgK:-25.(2SF)	0	ENS	773
3	25	1	Unknown	lgK:-22.4 (3SF)	4	EES	
4	25	1	Unknown	lgK:-23.4 (3SF)	0	ENS	773

Reaction No. 67539							
DETAP-4 + Pm+3 = Pm+3_DETAP-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:15.1 (3SF)	3	MGL	6529

Reaction No. 67540							
Pm+3 + H+1_DETAP-4 = Pm+3_H+1_DETAP-4							

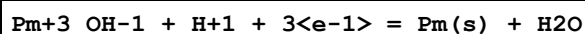
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:9.5 (2SF)	3	MGL	6529

Reaction No. 71040



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.2 (2SF)	4	CRV	6330
2	25	0	Inf. Dilution	E:-2.2 (2SF)	3	CRV	7761
3	25	0	Inf. Dilution	dH:407.3 (4SF) kJ	3	CRV	7761

Reaction No. 72057



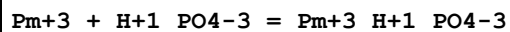
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-108.0 (4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-2.14 (3SF)	0	CRV	7761

Reaction No. 72059



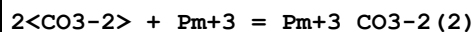
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-140.2 (4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-2.78 (3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:719.2 (4SF) kJ	3	CRV	7761

Reaction No. 72195



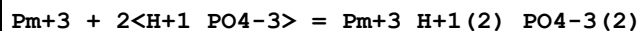
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.27 (3SF)	3	CRV	8781

Reaction No. 72206



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.08 (4SF)	4	EFE	8712
2	25	0	Inf. Dilution	lgK:11.96 (4SF)	3	CRV	8781

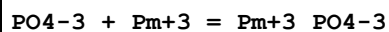
Reaction No. 72211



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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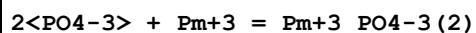
1	25	0	Inf. Dilution	lgK:8.81(3SF)	3	CRV	8781
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Reaction No. 72283



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.95(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:11.95(4SF)	4	EFE	8712

Reaction No. 72286



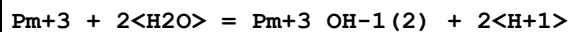
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.96(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:19.96(4SF)	4	EFE	8712

Reaction No. 72325



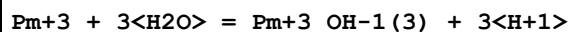
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.06(3SF)	4	EFE	8712

Reaction No. 72326



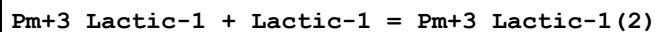
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-16.78(4SF)	4	EFE	8712

Reaction No. 72327



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-26.15(4SF)	4	EFE	8712

Reaction No. 75923



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	1.5	KCl	lgK:1.70(3SF)	5	MEP	11516

Note (1): Sakanoue M et al., Bull. Chem. Soc. Jpn., 1972, 45, 3429

2	20	0.2	NaClO4	lgK:1.77(3SF)	5	MIX	11516
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Note (2): Winkler K et al., Z. Phys. Chem., 1968, 238, 348

Reaction No. 78064

PO4-3 + Pm+3 + 2<H+1> = Pm+3_H+1(2)_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.96(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:22.03(4SF)	4	CRV	32224

Reaction No. 78374							
Citric-3 + Pm+3 + H+1 = Pm+3_H+1_Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.05(4SF)	1	ELC	

Reaction No. 78397							
2<Citric-3> + Pm+3 + H+1 = Pm+3_H+1_Citric-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.57(4SF)	1	ELC	

Reaction No. 78415							
Gly-1 + Pm+3 + H+1 = Pm+3_H+1_Gly-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.46(4SF)	1	ELC	

Reaction No. 78422							
2<Citric-3> + Pm+3 + 2<H+1> = Pm+3_H+1(2)_Citric-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.22(4SF)	1	ELC	

Reaction No. 79222							
PO4-3 + Pm+3 + H+1 = Pm+3_H+1_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.6(4SF)	1	ELC	

Reaction No. 79233							
2<PO4-3> + Pm+3 + 2<H+1> = Pm+3_H+1(2)_PO4-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.46(4SF)	1	ELC	

Reaction No. 80249							
1.5<O2(g)> + 2<Pm(s)> = Pm2O3(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	dG:-1736.(4SF)kJ	5	CRV	30332
Note (1): Value from LW3JTH option 2							
2	25	0	Inf. Dilution	dH:-1824.2(5SF)kJ	5	CRV	30332

Reaction No. 80715							
$\text{Pm}+2 + \text{H}+1 + 0.25\langle\text{O}_2\rangle = \text{Pm}+3 + 0.5\langle\text{H}_2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:63.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:65.28(4SF)	0	CRV	31585

Reaction No. 81555							
$\text{Pm}+3_{\text{OH}-1(3)}(\text{s}) + 3\langle\text{H}+1\rangle = \text{Pm}+3 + 3\langle\text{H}_2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.0(3SF)	1	ELC	

Reaction No. 81556							
$\text{Pm}_2\text{O}_3(\text{s}) + 6\langle\text{H}+1\rangle = 2\langle\text{Pm}+3\rangle + 3\langle\text{H}_2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:52.9(3SF)	1	ELC	

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CIU	Critical IUPAC publication (crit.)
CMN	Several experimental values (crit.)
CRV	Critical review
EDH	Debye-Hueckel corrections
EES	Personal estimate
EFE	Free Energy relations
EIE	Interpolation/Extrapolation (not Debye-Hueckel)
EIU	Critical IUPAC review (estim.)
ELC	Linear Combination of Reactions
ENS	Not specified
EOD	Calculated from data of others
EOR	Other Reaction Constants
ESC	Standard conditions anchor
MCE	Cation exchange
MDG	Combination of Thermodynamic Data
MDS	Distribution between two phases
MEP	Electrophoresis
MGL	Glass electrode
MIX	Ion exchange
MLC	Ligand competition
MSC	Miscellaneous
MSL	Solubility
MTD	Temperature difference
MTP	Migration or transference number

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